



Diffusion of complex contagions is shaped by a trade-off between reach and reinforcement

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How does social network structure amplify or stifle behavior diffusion? Existing theory suggests that when social reinforcement makes the adoption of behavior more likely, it should spread more—both farther and faster—on clustered networks with redundant ties. Conversely, if adoption does not benefit from social reinforcement, it should spread more on random networks which avoid such redundancies. We develop a model of behavior diffusion with tunable probabilistic adoption and social reinforcement parameters to systematically evaluate the conditions under which clustered networks spread behavior better than random networks. Using simulations and analytical methods, we identify precise boundaries in the parameter space where one network type outperforms the other or they perform equally. We find that, in most cases, random networks spread behavior as far or farther than clustered networks, even when social reinforcement increases adoption. Although we find that probabilistic, socially reinforced behaviors can spread farther on clustered networks in some cases, this is not the dominant pattern. Clustered networks are even less advantageous when individuals remain influential for longer after adopting, have more neighbors, or need more neighbors before social reinforcement takes effect. Under such conditions, clustering tends to help only when adoption is nearly deterministic, which is not representative of socially reinforced behaviors more generally. Clustered networks outperform random networks by a 5% margin in only 22% of the parameter space under its most favorable conditions. This pattern reflects a fundamental trade-off: Random ties enhance reach, while clustered ties enhance social reinforcement.

diffusion of behaviors | complex contagions | social influence | agent-based modeling

Social network structure has been shown to be consequential in the diffusion of behaviors, information, or ideas (1–3). The question of how network structure can amplify or stifle the spread of behavior has important implications across a wide range of fields, including health (4, 5), sociology (1), economics (6), organization science (7), and marketing (8). When behaviors are sensitive to the number of influential neighbors due to strategic complementarity, social pressure, credibility, or legitimacy (9, 10), they can spread better—both faster and farther—on clustered networks. On clustered networks, neighbors of a given individual tend to be themselves densely connected, enabling repeated exposure to multiple adopting neighbors who serve as a source of social reinforcement, increasing the likelihood of adoption. Conversely, some behaviors are not sensitive to social reinforcement. Instead, the chance of adoption is independent of how many different adopting neighbors an individual is exposed to. This is similar to the spread of biological contagions where contact with a single infected individual is sufficient to catch the disease (11, 12). When behaviors are not sensitive to social reinforcement, clustering merely adds unnecessary redundancy—“wasting” exposures on people who have already adopted—thus slowing down spread and reducing its reach (2). Socially reinforced spreading processes have been called “complex contagions” whereas processes not sensitive to social reinforcement have been called “simple contagions” (10).

Important parts of the prior literature on diffusion have cast this distinction about which behavior spreads better on which network as a sharp dichotomy: Clustering facilitates the spread of socially reinforced behaviors but hinders the spread of behaviors that do not benefit from social reinforcement (3). Recent work, however, suggests this dichotomy is an artifact of modeling behavior as deterministic (13, 14). They find that relaxing the assumption of deterministic adoption challenges the idea that socially reinforced behavior benefits from redundant—but not random—ties. By allowing for small amounts of probabilistic adoption (but keeping other parts deterministic), they show that socially reinforced behaviors spread faster on random networks just like

Significance

A dominant theory about the diffusion of behaviors on networks suggests that when the adoption of a behavior is more likely with social reinforcement, it will spread more on clustered networks. By contrast, when the behavior does not benefit from social reinforcement, it will spread more on random networks. We develop a general model of social reinforcement that encompasses probabilistic adoption and find this dichotomy to break down. We find many cases of socially reinforced behaviors that spread better on random networks. Socially reinforced behaviors that do spread better on clustered networks approach deterministic spreading processes that are unrepresentative of the probabilistic nature of social contagion born out in empirical research.

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nonsocially reinforced behaviors. Despite this progress, it remains unclear how general the “faster on clustered networks/slower on random network” dichotomy is when adoption is consistently modeled probabilistically, which is both more realistic and reflective of the empirical literature on peer influence (15–20). This is particularly important since some empirical (21) and theoretical studies (3) incorporating probabilistic adoption do find evidence for socially reinforced behaviors spreading farther and faster on clustered networks, apparently contradicting the results in refs. 13 and 14. So why, and under what conditions, is clustering beneficial to the spread of behaviors when the behavior benefits from social reinforcement and is consistently modeled with probabilistic adoption? Here, we show that the dichotomy—and the conflicting results—can be explained by a fundamental trade-off between the benefit of redundancy that social reinforcement may harness (a reinforcement effect) and the loss that this same redundancy causes by limiting reach to fewer individuals (a reach effect). Whether clustering is harmful or helpful depends on the precise strength of both effects. When the benefits of reinforcement outweigh the loss in reach, socially reinforced behaviors will spread better on clustered networks. When the reverse is true, random networks will be more beneficial.

As one way to quantify the trade-off between reinforcement and reach, we develop a threshold model of homogeneous behavior adoption with tunable probabilistic adoption. Socially reinforced behaviors have often been modeled with deterministic threshold models (3, 22, 23), where exposure to a specified threshold number of influential neighbors is required for adoption. That is, adoption is impossible below the threshold, but guaranteed above. When adoption is instead modeled probabilistically, surpassing the threshold of neighbors allows social reinforcement to “kick in,” increasing the likelihood of adoption from the below threshold base rate. Adoption becomes possible below the threshold, yet is not guaranteed above. Our model relies on two parameters to incorporate probabilistic adoption: One tunes the base rate of adoption (below the threshold); the other tunes the socially reinforced adoption rate (above the threshold). Each parameter ranges from 0—implying no adoption above (below) the threshold—to 1—implying guaranteed deterministic adoption above (below) the threshold. This model thus efficiently captures diffusion processes in the space from simple to complex contagions.

In this paper, we use agent-based modeling and analytical techniques to analyze diffusion processes where the socially reinforced adoption rate equals or exceeds the base rate of adoption (i.e., the scenarios in which earlier literature had suggested that clustered networks should be superior at facilitating spread compared to random networks; 24). Additionally, we consider the effect of other relevant features of the diffusion process and network structure including the threshold of neighbors an individual must be exposed to in order to adopt at the socially reinforced above threshold rate (“social reinforcement threshold”), how long an individual remains influential after adopting (“time of influence”), and the number of neighbors each individual in the network has (“degree”). For each parameter combination, we compare diffusion on clustered ring lattice networks (2) to that of regular random networks constructed by rewiring clustered networks (25), while holding network size and degree constant across network types (we explore diverse variations in our robustness tests). As outcomes, we are interested in both the reach and speed of spread.

We find there are four qualitatively distinct regions in the space defined by our two parameters: 1) clustered networks

outspread random networks, 2) random networks outspread clustered networks, 3) full spread is reached on both network types but at a faster rate on random networks, or 4) there is minimal spread on both network types. This means that there are behaviors with significant social reinforcement that spread more on random networks, as well as those that spread more on clustered networks. The canonical result of “better spread on clustered networks” in fact represents only a small portion of the parameter space. This region shrinks even more when behaviors have long times of influence, high social reinforcement thresholds, and greater network degree. When time of influence is long, full spread dominates. Clustered networks outspread random ones only when below-threshold adoption is close to zero.

This paper makes the following contributions. First, our model explores a larger parameter space of simple and complex contagions, subsumes previous models as special cases, and better reflects the probabilistic nature of real world human behavior. Previous work that numerically simulates models with both probabilistic base rate and socially reinforced adoption did not incorporate thresholds (26–28), and work using threshold models included probabilistic base rate adoption (13, 14) or probabilistic socially reinforced adoption (3), but not both.* This integrative modeling framework unifies several strands of the prior literature and strengthens the consensus on the role of social reinforcement by highlighting large regions where behaviors with probabilistic adoption spread faster on random networks despite social reinforcement (aligning with the conclusions of refs. 13 and 14) as well as regions in which socially reinforced behavior spreads better on clustered networks (supporting the conclusions of refs. 3 and 21). Our findings clarify that social reinforcement is a necessary but insufficient condition for clustering to be beneficial to the diffusion of behaviors.

Second, the effect of clustering is not exclusively helpful (3, 21) or harmful (13) to the diffusion of socially reinforced behaviors with probabilistic adoption as previous work suggests.† Rather, behaviors are subject to a trade-off between reach (from random ties) and reinforcement (from clustered ties). Despite its simple nature, our model offers a lens that brings together diverse studies that variously show the benefits of both random and clustered ties.

Third, we systematically explore time of influence which few studies have varied as a parameter of interest, given its challenging analytical tractability.‡ By more systematically exploring time of influence, we arrive at contradictory results compared to prior studies (14) that find time of influence only stifles spread on random networks but not on clustered networks. Instead, we find that shorter times of influence hinder spread on both network types. This increases Region 1 (where clustered networks outspread random networks) and Region 2 (where random networks outspread clustered networks) at the expense of Region 3 where behaviors can spread on both network types.

Fourth, by casting the benefits of clustering as a trade-off between reach and reinforcement, we can quantify how the regions in which clustered networks (as opposed to random networks) better spread a behavior expand or contract depending on other modeling parameters. Across a wide range of parameter

*Eckles et al. (13) theoretically model diffusion processes with both probabilistic base rate adoption and social reinforcement, but only numerically simulate one case when base rate adoption is above 0 and the socially reinforced adoption rate is set to 0.5.

†But see Sassine and Rahmandad (14).

‡The closest is the “sender inactivation” parameter in ref. 14 and one numerical robustness check each in refs. 3 and 13.

values, socially reinforced behaviors spread along both random and clustered ties. Even small increases in how long individuals remain influential after adopting, how many neighbors an individual has, or how many neighbors are required to adopt at the socially reinforced rate all increase the region where full spread is reached on both network types (though at a faster rate on random networks) and decrease the region where clustered networks are advantageous. Clustered networks only outperform random networks among behaviors with increasingly deterministic adoption, (when the base rate is close to 0 and the socially reinforced rate is close to 1) which are not representative of probabilistic complex contagions as a whole.

In summary, we find that prior work has somewhat overestimated the benefits of clustering for the spread of socially reinforced behaviors (3). Whereas the results reported in ref. 21 are possible under certain conditions (and fully consistent with the complex contagions spread faster on clustered networks claim) they do not represent the dominant pattern. For most cases, random networks spread a behavior equally or better. This suggests that greater diffusion on clustered networks is not a defining feature of complex contagions as a whole.

Establishing Microfoundations of Social Influence: Modeling Behaviors with Probabilistic Adoption. We introduce a model that describes the microlevel process of social influence, formalizing the differences between socially reinforced complex contagions and nonsocially reinforced simple contagions. All individuals in the network begin having not adopted a behavior (they are “susceptible”), except for several randomly chosen “seed” individuals who have already adopted and can influence their immediate neighbors to adopt (the seeds are “infected” individuals; Fig. 1A). Those who have adopted a behavior remain influential toward their neighbors for a set time length (T) after which they can no longer influence others (they are “recovered”). This mirrors the Susceptible-Infective-Recovered (SIR) model from epidemiology (12, 29).

For each time step, all susceptible individuals are simultaneously exposed to any neighboring individuals who are currently influential. With every exposure to an influential neighbor, an

individual may adopt the behavior with a certain “per-exposure” probability. This per-exposure probability of adoption is defined by $p(c)$, where c indexes the number of different influential neighbors an individual has been exposed to from the start of the simulation.

$$p(c) = \begin{cases} 0, & \text{if } c = 0 \\ p_1, & \text{if } 1 \leq c < i \\ p_2, & \text{if } c \geq i. \end{cases}$$

All individuals follow this adoption rule identically and there is no heterogeneity among individuals except for network position.[§]

When an individual is not in contact with any influential neighbors, they cannot adopt the behavior. If an individual has been exposed to less than i different influential neighbors, they will adopt with the base rate adoption probability of p_1 . If the number of different adopting neighbors an individual is exposed to equals or exceeds i , the social reinforcement threshold, an individual adopts the behavior with the socially reinforced probability p_2 . In practice, even if an individual is exposed to multiple neighbors within one time step, the number of exposures is still counted serially. For instance, if $i = 2$ and an unexposed individual is exposed to three influential neighbors for the first time within one time step, one neighbor “transmits” the behavior with the base rate probability of p_1 while the other two transmit the behavior with the socially reinforced probability of p_2 . The difference between p_1 and p_2 quantifies the amount of social reinforcement the adoption of a behavior is sensitive to, with the idea being that multiple exposures reinforce the likelihood of adoption beyond that of the base rate adoption probability p_1 .

Setting different values of p_1 and p_2 can parameterize behaviors with different levels of base rate and socially reinforced adoption rates. This allows us to recover well studied forms of complex and simple contagions, while at the same time allows us to examine overlooked regions of the space (Fig. 1C). When $p_1 = p_2$, the threshold parameter i has no effect and $p(c)$ remains constant across all additional contacts c . Increasing

[§]See SI Appendix, section I for extensions with heterogeneous adoption.

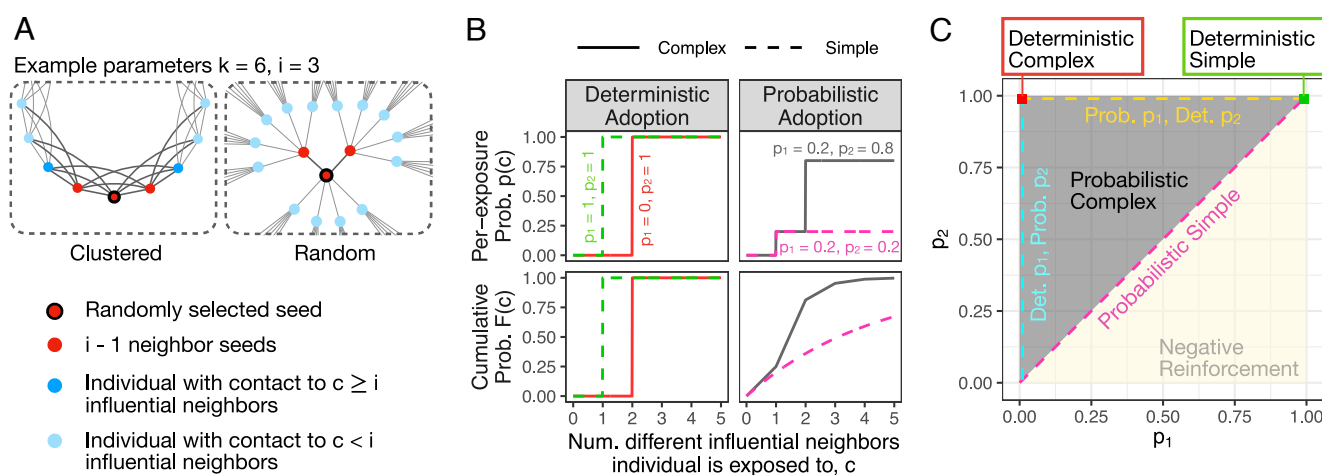


Fig. 1. Micro and macro views of diffusion on clustered and random networks. (A) The seeding structure and early time diffusion in random and clustered networks, with example parameters $k = 6$ and $i = 3$. Both network types are seeded by selecting one random individual and $i - 1$ neighbors. Random networks reach more individuals ($ki - 2i + 2$ in the first time step), but all adopt at the lower, base rate p_1 . Clustered networks reach less individuals (k in the first time step for this example, a different set of $i - 1$ neighbors could be chosen) but more individuals receive reinforcing signals, enabling adoption at the higher p_2 rate. This illuminates a trade-off in having more or less redundant ties. (B) Per-exposure and cumulative adoption probabilities for simple and complex contagions with deterministic and probabilistic adoption. Example parameters $p_1 = 0.2$ and $p_2 = 0.8$ are used for behaviors with probabilistic adoption. (C) The space of all possible p_1, p_2 values that uniquely define a behavior, or adoption trajectory ($p(c), F(c)$). Our analysis focuses on $p_1 \leq p_2$.

the number of influential neighbors an individual is exposed to does not increase an individual's per-exposure probability of adoption, so the behavior is considered a simple contagion. The behavior is a simple contagion with deterministic adoption when $p_1 = p_2 = 1$ (or "deterministic simple contagion"), and a simple contagion with probabilistic adoption (or "probabilistic simple contagion") when $0 < p_1 = p_2 < 1$. When $p_1 \neq p_2$, the behavior is sensitive to social reinforcement, so it is a complex contagion. Social reinforcement can be positive when the per-exposure adoption probability increases with exposure to multiple influential neighbors, $p_1 < p_2$ (as theorized in complex contagion about costly behaviors such as attending a protest) or negative if exposure to additional influential neighbors somehow dampen each other, $p_1 > p_2$ (e.g., spreading a rumor may become less satisfying if many people already know it). Under both positive and negative social reinforcement, the complex contagion can have deterministic adoption ($p_1 = 0, p_2 = 1$ in the positive case; $p_1 = 1, p_2 = 0$ in the negative case) or probabilistic adoption ($0 \leq p_1 < p_2 \leq 1$ but not including $p_1 = 0, p_2 = 1$ or $p_1 = 1, p_2 = 0$). We focus on simple contagions and complex contagions with positive social reinforcement with deterministic or probabilistic adoption, where $p_1 \leq p_2$.

Among complex contagions with positive social reinforcement, the social reinforcement threshold i models how many different neighbors an individual must be in contact with in order to adopt at p_2 instead of p_1 , "activating" this positive reinforcement effect. Holding constant the total number of neighbors an individual has, or degree k , while increasing i increases the costliness of adopting a behavior. Contact with more socially reinforcing neighbors relative to the total number of neighbors is required to adopt at the higher, socially reinforced rate p_2 . This is similar to other threshold models explored in the literature (22, 23, 30) where individuals adopt a behavior based on whether a certain threshold of neighbors adopt. However, rather than governing deterministic adoption, surpassing i only increases the likelihood of adoption from p_1 to p_2 . As we are interested in providing a minimal model that systematically varies adoption and social reinforcement, we model adoption in probabilistic terms while retaining a homogeneous social reinforcement threshold i that serves as a model parameter.

The length of time an individual remains influential after adopting, or what we refer to as "time of influence" T , models a distinction between behaviors that remain transmissible for longer periods of time as opposed to shorter periods of time. For instance, behaviors that remain highly visible, salient, or relevant over time (such as changing a highly visible profile picture on social media) may exhibit longer times of influence compared to behaviors where visibility quickly diminishes with time (such as liking a post that quickly becomes buried under newer content).[‡] At the extreme, such a distinction between diffusion processes with longer or shorter times of influence is analogous to the differences between the Susceptible-Infective (SI) model, where the time of influence is infinite (individuals remain influential for the entire simulation after adopting), and SIR model, where the time of influence is some finite value. It is well understood that SI and SIR models exhibit different diffusion patterns, suggesting that variation in time of influence may significantly affect how a behavior spreads (12, 29, 32).

Core to understanding the difference between simple and complex contagions is making a distinction between gains in diffusion from social reinforcement on the one hand, and gains

from receiving repeated exposures to influential neighbors on the other. The former, benefiting from social reinforcement, refers to an increase in the per-exposure probability of adoption of a behavior dependent on an increase in the number of different influential neighbors an individual is exposed to (adopting at p_2 instead of p_1). This is characteristic of complex contagions with positive reinforcement studied here. The latter, benefiting from repeated exposures to influential neighbors, refers to the extent to which the cumulative probability increases with more exposures, simply from the nature of probability (the chance of observing at least one coin toss to come up heads is higher when we flip two coins than when flipping just one [i.e., $p(\text{at least one head}) = 1 - (1 - 0.5)^2 = 0.75$]). While benefiting from social reinforcement is only possible when the number of different influential neighbors exceeds the threshold i , benefiting from redundant exposures occurs with every exposure, regardless of whether they are from the same neighbor or different neighbors. Nonsocially reinforced simple contagions with probabilistic adoption benefit only from increasing exposure to influential neighbors, but complex contagions with positive social reinforcement benefit from both redundant exposure to influential neighbors and the socially amplified adoption probability p_2 (when exposures exceed the threshold i).

This difference can be formalized by the cumulative probability $F(c)$ of the per-exposure probability of adoption $p(c)$, where $F_C(c) = P(C \leq c)$ (Fig. 1B). The cumulative probability of adopting a simple contagion can be expressed as, $F(c) = 1 - (1 - \beta)^c$, where $p_1 = p_2 = \beta$. When the behavior is a simple contagion with deterministic adoption, $p_1 = p_2 = 1$, $F(c) = 1$, and the likelihood of adopting the behavior does not increase with additional exposures after the first exposure since the behavior is guaranteed to be adopted after the first exposure. However, when $0 < \beta < 1$ and the behavior is a simple contagion with probabilistic adoption, $F(c)$ increases with additional exposures to influential neighbors, similarly to that of complex contagions, even though the behavior is not more likely to be adopted with social reinforcement. Though in the case of complex contagions with probabilistic adoption, $F(c)$ increases at a faster rate compared to simple contagions with probabilistic adoption with the same base rate probability p_1 . This is visible in the *Bottom Right* panel of Fig. 1B: While the simple contagion with probabilistic adoption experiences an increasing cumulative adoption probability from exposure to more influential neighbors (albeit with diminishing returns), the increase for the complex contagion with probabilistic adoption is higher.

Given that both simple and complex contagions with probabilistic adoption benefit from repeated exposures through redundant ties characteristic of clustering, but can also be transmitted along nonredundant, random ties that can quickly reach disparate parts of the network, it becomes theoretically ambiguous as to whether the presence of clustering would be beneficial for spread in either case. Probabilistic simple contagions benefit from redundant exposures, while probabilistic complex contagions with nonzero p_1 can benefit from reaching more unique individuals even through nonredundant ties (Fig. 1B). This stands in contrast to the clean cut deterministic case where complex contagions spread better on clustered networks because they only benefit from redundant ties, and simple contagions spread better on random networks because they only benefit from nonredundant ties. The relative strengths of these two effects, gains from redundant ties as opposed to gains from nonredundant ties, will determine which network spreads behavior "better." By exploring this model, we will show that socially reinforced complex contagions spread farther and faster on clustered

[‡]This is similar to incorporating memory or activation time parameters into a contagion (11, 14, 31).

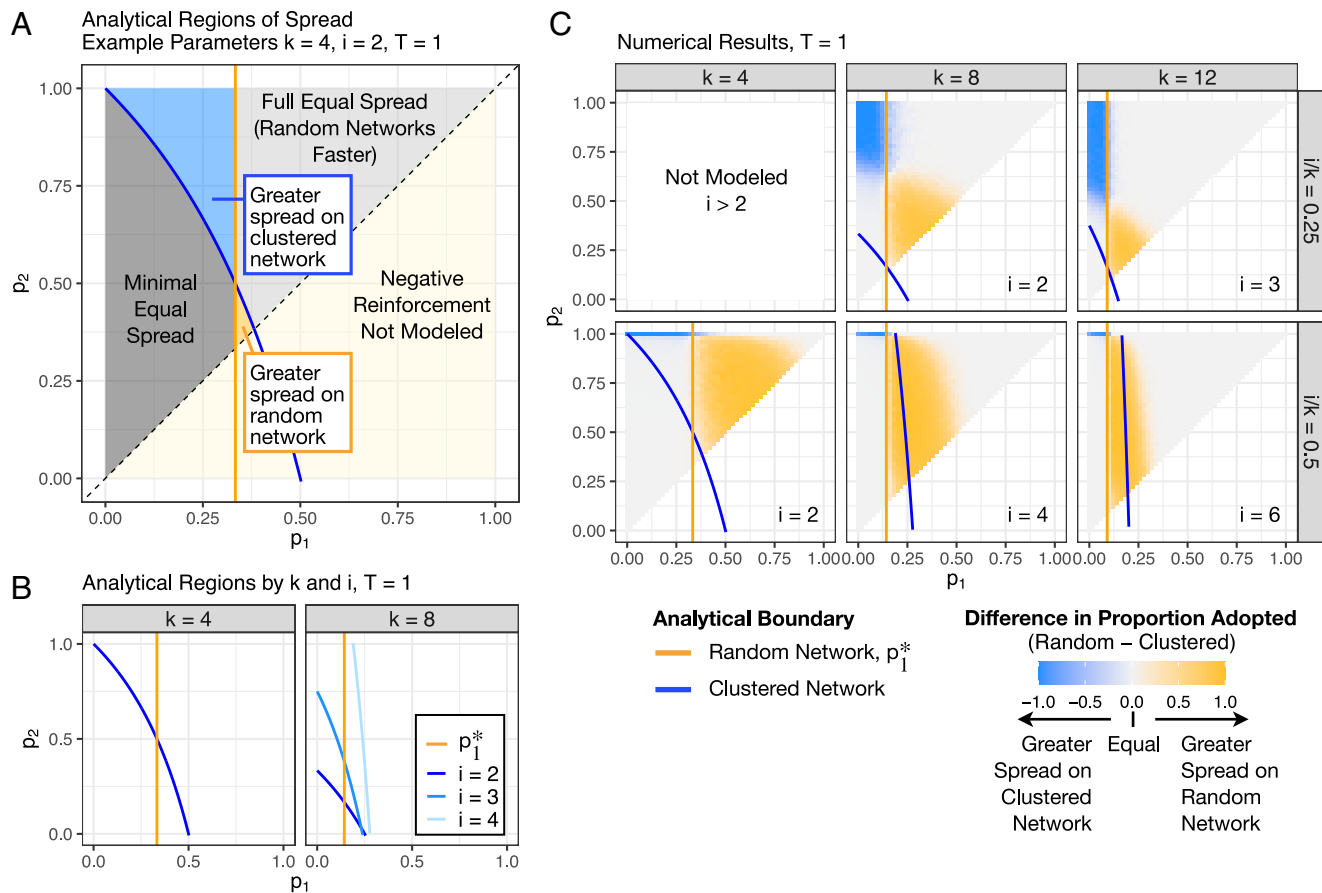


Fig. 2. Regions of spread on clustered and random networks. (A) Analytical regions of spread on random and clustered networks with example parameters $k = 4, i = 2, T = 1$. The blue and orange lines denote where full spread is likely to be reached on either clustered or random networks respectively, demarcating four distinct regions in the $p_1 \leq p_2$ space: clustered networks outspread random networks (blue), clustered networks outspread random networks (orange), full equal spread on both networks with random networks faster (light gray), and minimal equal spread on both network types (dark gray). (B) Analytical boundaries for different degree k and social reinforcement threshold i . (C) Analytical boundaries (colored lines) overlaid on simulation results (shaded regions, proportion of final adopters averaged over 100 trials) for different k and i . Random networks have a greater proportion of adopters in the orange region, clustered networks have a greater proportion of adopters in the blue region. The analytical boundaries closely map to numerical results on random networks and underestimate values of p_1 and p_2 where full spread is reached on clustered networks.

networks only in a small area of the $p_1 \leq p_2$ parameter space, whereas in the majority of the parameter space the random network either performs equally or better. We additionally test the effects of differing degree (k), social reinforcement threshold (i), and time of influence (T ; see *Materials and Methods*).

Results

We observe a large amount of heterogeneity in how behaviors in the $p_1 \leq p_2$ space diffuse on clustered and random networks. Rather than clustering being universally helpful or harmful to spread, results show four qualitatively distinct regions: 1) clustered networks outspread random networks, 2) random networks outspread clustered networks, 3) full spread is reached on both network types but at a faster rate on random networks, or 4) minimal spread on both network types. This variation is entirely explained by the extent to which the base rate of adoption (p_1), socially reinforced adoption (p_2), network degree (k), time of influence (T), and social reinforcement threshold (i) modulate the trade-off between the benefits of reach from random ties, and the benefits of social reinforcement from clustered ties. We demonstrate this both with simulations and a simplified analytical proof (Fig. 2; see *Materials and Methods*).

On random networks, the majority of parameter combinations will either spread readily or not spread at all (*SI Appendix, Fig. S6*). This creates an observable boundary in the $p_1 \leq p_2$ space where the likelihood of reaching full spread quickly transitions from very low to very high. This transition occurs when the base rate adoption p_1 is greater than a threshold value p_1^* , such that $p_1^* = 1 - (1 - (1/(k - 1)))^{1/T}$. If p_1 crosses this threshold, on average every adopting individual can influence at least one neighbor, sustaining the diffusion process. This reflects the logic of an epidemic threshold (29, 32). As either network degree or time of influence increases, the chance that an adopting individual may influence their neighbor increases. Individuals have more chances to transmit behavior, either by having more neighbors or by having more repeated transmission as the time of influence increases. On the macrolevel, this decreases p_1^* , and increases the area within the $p_1 \leq p_2$ space where a behavior can attain full spread on a random network.

Spread on clustered networks is also characterized by a sharp transition from regions where spread is highly unlikely to a region where attaining full spread is highly likely. Overcoming this transition to a region where full spread is likely on clustered networks depends on either having a high enough p_1 such that a behavior can spread without the benefit of social reinforcement from redundant ties, or a high enough p_2 such that the social

reinforcement from redundant ties alone can continue the diffusion process. When p_1 is too low, a behavior cannot be diffused along nonredundant ties. Instead, the diffusion process must be sustained along clustered ties by a high p_2 . As p_1 increases, behavior can spread more readily through both redundant and nonredundant ties, and can thus diffuse even with lower levels of p_2 . Once p_1 is high enough that a diffusion process could be sustained on clustered networks without added social reinforcement from redundant ties, full spread can be reached regardless of the value of p_2 . By the same mechanism as spread on random networks. Increasing degree and time of influence both facilitate spread on clustered networks, similar to that of random networks. As a result, the region where behaviors can spread on clustered networks expands to encompass lower values of p_1 and p_2 . Increasing i requires contact with a greater number of adopting neighbors before social reinforcement can “kick in.” This makes full spread on clustered networks less likely and increases the required values of p_1 and p_2 for diffusion to occur.

Combining the forces driving spread on random and clustered networks reveals four distinct regions in the $p_1 \leq p_2$ space (Fig. 2 A and C). In Region 1, clustered networks outspread random networks. With higher p_2 but lower p_1 , behaviors can spread through socially reinforcing redundant ties, but not random ties. In Region 2, behaviors spread on random networks, but not clustered networks. Behaviors have a high enough p_1 to overcome the p_1^* threshold, spreading readily on random networks. However, neither p_1 nor p_2 is high enough for behaviors to spread on clustered networks. In Region 3, p_1 and p_2 are both high enough that behaviors can fully diffuse on either random or clustered networks. That being said, the ability of random ties to reach many individuals through short paths (when few individuals serve as intermediaries between any two individuals in the network), means that behaviors reach full spread at a faster rate on random networks compared to clustered networks in this region. Finally in Region 4, both p_1 and p_2 are low, so there is minimal spread on both network types. Neither the redundant ties of clustered networks nor the ability for random networks to reach many unique individuals

can be used advantageously. As random networks will always diffuse a behavior faster in Regions 2 and 3, focusing on each of these regions can describe both the speed and reach of spread. Our analytical model maps closely to the boundary where full spread is reached on random networks, and serves as an informative estimate for where full spread on clustered networks is reached (SI Appendix, Fig. S1). Moreover, this analytical model accurately reflects how variables including network degree, time of influence, and the social reinforcement threshold affect spread.

Altogether, Region 1 where clustered networks outperform random networks constitutes a minority of the $p_1 \leq p_2$ space that approaches a deterministic setting (low p_1 , high p_2). Region 1 is maximized and encompasses more probabilistic regions of the space when time of influence is short, degree is low, and the social reinforcement threshold is low. This most likely reflects the parameter settings characteristic of the empirical behaviors studied in ref. 21 which found faster spread on clustered networks (for a more detailed comparison see SI Appendix, section L). Even under these ideal conditions, across all the parameter combinations we examine, clustered networks spread a behavior to five percent more of the total network compared to random networks for at most only 22 percent of the $p_1 \leq p_2$ space (for the parameter combination $k = 10$, $T = 1$, $i = 2$; Fig. 3A).[#] The minimal amount of social reinforcement (ratio of p_2 to p_1 , p_2/p_1) we observe where clustered networks spread a behavior to at least five percent more of the network compared to random networks occurs when p_2 is nearly two times that of p_1 ($p_2/p_1 = 1.90$; SI Appendix, Fig. S4). This is important insofar that among empirical studies of peer influence, it is rarely the case that the ratio of socially reinforced to nonsocially reinforced adoption rates ever exceed two (see refs. 13 and 19 for reviews).

[#]We select a difference in spread of five percent of the total individuals in the network as a threshold for substantive significance. For instance, on networks with $k = 4$ and 1,000 individuals this would be a difference of 50 individuals. We additionally test for statistical differences using a nonparametric Kolmogorov–Smirnov(KS) test. Although a KS test slightly increases the area where clustered networks outspread random networks, this region still constitutes a minority of the total space and shrinks with greater degree, time of influence, and social reinforcement threshold. See SI Appendix, section C.

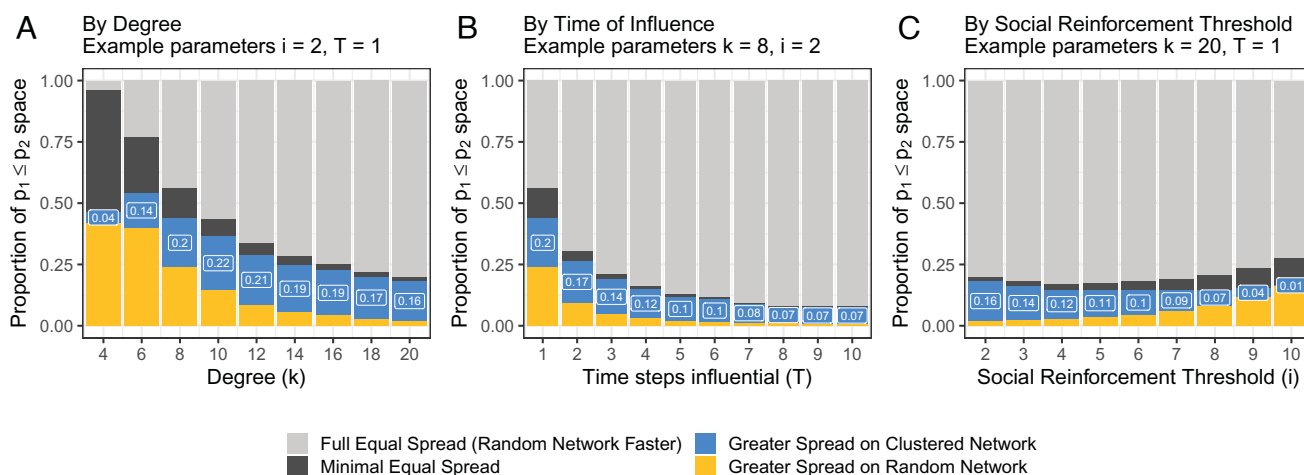


Fig. 3. Effect of degree, time of influence, and social reinforcement threshold on regions of spread in the $p_1 \leq p_2$ space. The proportion of the $p_1 \leq p_2$ where clustered networks outperform random networks (blue), random networks outperform clustered networks (orange), both network types have equal minimal spread (dark gray), or both network types have equal full spread (but faster on random networks, light gray) for varying degree (A), time of influence (B), and social reinforcement threshold (C). Spread, or the proportion of adopters at the end of the simulation, is averaged over 100 simulations of the same parameter combination and one network is considered to outperform another if there is at least a five percent difference in the proportion of adopters. Full equal spread is where the difference in the proportion of adopters between the two network types is within five percent and at least 60 percent of the network adopts the behavior for both network types. Minimal equal spread is where the difference in the proportion of adopters between the two network types is within five percent and less than 60 percent of the network adopts the behavior for either network type.

With the exception of networks with low degree where the substantive difference in spread between the two network types is smaller ($k = 4, 6, 8$), greater degree, time of influence, and social reinforcement all shrink Region 1 toward more deterministic areas of the $p_1 \leq p_2$ space (Fig. 3). First, increasing the social reinforcement threshold raises the number of different neighbors an individual must be exposed to in order to adopt at the p_2 rate, requiring a p_2 closer to 1 to sustain diffusion on clustered networks. This shrinks Region 1, making it more deterministic. Second, interventions that increase spreading on both clustered and random networks (such as greater degree or longer time of influence) decrease p_1^* on random networks and enable spread on clustered networks for lower p_1 and p_2 values. As a result, Region 3 where both network types reach full spread, though at a faster rate on random networks, expands at the expense of Regions 1 (clustered networks spread behaviors better) and 2 (random networks spread behaviors better; Fig. 2 and *SI Appendix, Fig. S6*). Region 1 narrows to smaller, more deterministic values of p_1 , but also extends to lower values of p_2 .

Since the shrinkage from smaller p_1 outpaces the gains with p_2 , Region 1 shrinks with greater degree and time of influence. By extending time of influence so that adopters remain influential for the entire simulation as modeled in refs. 3, 13, and 14, aside from where p_1 approaches 0 and $p_2 > 0$ where clustered networks perform best, Region 3 dominates (*SI Appendix, Fig. S6E*). This means for deterministic $p_1 = 0$, clustered networks outperform random networks for any amount of (deterministic or probabilistic) $p_2 > 0$, explaining the results of ref. 3 who use the point $p_1 = 0, p_2 = 0.5$ as a robustness check for probabilistic adoption (*SI Appendix, section K*). However, virtually any increase in p_1 from 0 would push a behavior into Region 3 where full spread is attained equally on both clustered and random networks, but at a much faster rate on random networks consistent with the core results of ref. 13.

Finally, our conclusions about the effects of time of influence diverge somewhat from those of ref. 14 who incorporate a similar parameter. Whereas they find that shorter times of influence advantage clustered networks (over random networks), they conclude this is because short times of influence stifle spread on random networks while clustered networks remain unaffected. Instead, we find that short times of influence limit spread on both network types, but for different parts of the $p_1 \leq p_2$ space. This insight derives from modeling all combinations of $p_1 \leq p_2$ instead of merely $0 \leq p_1 \leq 1, p_2 = 1$ as Sassine and Rahmandad (14) do. For instance, if we consider networks where $k = 4$, there is no difference in spread on clustered networks from $T = 1$ (*SI Appendix, Fig. S6A*) to a longer time of influence ($T = 5$) when $p_2 = 1$ (*SI Appendix, Fig. S6C*). However, there is a dramatic change in spread from $T = 1$ to $T = 5$, when $p_2 < 1$.

Discussion

Over the past 15 y, the theory of complex contagion has emerged as one of the most influential and paradigm-shifting ideas in the study of social influence (10). In contrast to the long-standing belief in the structural advantages of random ties for spreading information or behaviors quickly and widely (1, 2), the complex contagion argument introduced a fundamentally different perspective. It posited that the adoption of costly behaviors—those requiring socially reinforcing signals from multiple neighbors—is more effectively supported by the redundant ties found in clustered networks (3). Whereas the theory was originally developed in a deterministic context, it assumed that

such findings would generalize to a probabilistic setting which more accurately captures the capriciousness of human behavior.

We show that the apparent dichotomy about which behavior spreads faster on which network can be explained by a fundamental trade-off between a reinforcement effect and a reach effect. Whether clustering is helpful or harmful to the spread of socially reinforced behavior depends on the precise strength of both effects. As a result, both results reported in the prior literature are possible: In one region of the parameter space between base rate and social reinforcement, behavior spreads better on clustered networks; in another region, behavior spreads better on random networks. The region in the parameter space where reinforcement outweighs reach is small, resembles the deterministic setting (where p_1 is close to 0 and p_2 close to 1), and unlikely to occur in real world diffusion processes. In the majority of cases, both networks spread a behavior equally far (though at a faster rate on random networks). We provide analytical bounds that divide the parameter space into regions in which each case occurs. Our numerical analyses additionally explore how the different regions expand or shrink when behaviors have long times of influence, high social reinforcement thresholds, and network degree is high. The general pattern is this: Once we allow for a shift from social reinforcement being necessary before adoption can occur to merely being amplified, diffusion processes cease to depend only on clustered ties and begin to benefit from random ties.

Our work builds on prior work, notably refs. 14 and 13 which made important contributions regarding stochastic diffusion processes. We add to the cumulative picture of what we know about complex and simple contagions. We quantify for which kinds of diffusion processes (in terms of base adoption rate and strength of social reinforcement) clustered as opposed to random networks spread a behavior better, how the different regions expand or shrink as other parameters change (degree, time of influence, and threshold), and provide important robustness tests (heterogeneity in adoption, one- and two-dimensional clustered networks, alternative seeding strategies, and intermediate levels of rewiring). We do so in a more systematic way, exploring both probabilistic above and below threshold adoption.

Regarding other models that have explored probabilistic spreading (e.g., ref. 3) we find that when adopters remain influential for the entire simulation, spread on clustered networks is sensitive to how close to deterministic the base rate adoption is rather than how deterministic above threshold adoption is. Some past work has thus overlooked the important effects of nondeterministic base rate adoption which results in faster spreading on random networks and not clustered networks.^{||} Regarding ref. 21, our synthetic replication reveals that there is indeed a region in the $p_1 \leq p_2$ space where socially reinforced behavior with probabilistic adoption spreads only on clustered networks. However, for many different levels of p_1 and p_2 there are also results showing spread exclusively on random networks, spread on both network types, or no spread at all. The study's choice of two-dimensional networks, networks with low degree, and the transmission of a behavior with a presumably low time of influence all serve to maximize the region where clustered networks are beneficial, increasing the likelihood of finding the result he did. The advantage of clustered networks is strongest when social reinforcement is necessary for adoption to occur, yet is substantially weaker when it merely amplifies adoption. In that case, random ties increase spread substantially.

^{||} Refs. 13 and 14 are notable exceptions.

At best, clustered networks only outperform random networks by at least five percent when social reinforcement is at least two times that of the baseline probability of adoption. Based on this, among empirical studies of peer influence or social contagion, it is rarely the case that the socially reinforced adoption rate is even proportionally high enough for clustered networks to outperform random networks (see refs. 13 and 19 for reviews). For most realistic, probabilistic human behavior, a random network will generally spread a behavior just as well, if not better, than a clustered network.

In summary, we develop a simple two-parameter model for simple and complex contagions that subsumes many prior models as special cases. We provide analytic bounds that demonstrate large amounts of heterogeneity in how well random and clustered networks diffuse socially reinforced behaviors. Our work bridges theory developed for deterministic and probabilistic threshold diffusion processes, painting a richer, cumulative picture of possible outcomes, resulting from a trade-off of reach (benefit of random ties) and reinforcement (benefit of clustered ties).

Our work is not without limitations. Whereas we explore a variety of different network structures, including one- and two-dimensional lattices and those with both clustering and short paths, future work could explore networks with heterogeneous degree (33, 34), or on empirical social networks. Our work is mostly focused on homogeneous individual behavior (although we have important robustness tests using heterogeneous adoption), and we do not examine potential correlations between clustering and other features that may affect diffusion including the relational strength of ties between individuals (35, 36), heterogeneity in the number of neighbors an individual has (33, 34, 37), or homophily of individual traits (38–41). All would be fruitful directions for future work.

Materials and Methods

Network Structure. In all networks, individuals have the same degree k . The number of individuals in the network is scaled to $n = 250k$ to preserve network density and isolate the effect of degree. Networks with $k = 4$ have 1,000 individuals, those with $k = 8$ have 2,000 individuals, and so on. Clustered networks are Watts–Strogatz style ring lattices (2) where each node is connected to their k nearest neighbors. Random networks are generated by taking the original ring-lattice and performing a series of degree-preserving edge swaps according to the procedure outlined in ref. 25 and implemented in networkX. Specifically, two random edges, say (u, v) and (x, y) , are selected and swapped to become (u, x) and (v, y) where the number of swaps is equal to the number of edges. If the edge swap disconnects the network, or if the new edge created with the swap already exists, the swap is rejected. For computational efficiency, the algorithm only checks for connectedness after every three swaps. This procedure uniformly samples the space of connected regular networks with a specified degree sequence (25). The resulting networks are regular random networks where all individuals have the same number of neighbors. This is similar to procedures used in related work (3, 14, 21, 28).

Adoption Rule. Individuals adopt the diffusing behavior with either 0, p_1 , or p_2 probability based on whether the number of influential neighbors they are in contact with, c , has overcome the social reinforcement threshold i .

$$p(c) = \begin{cases} 0, & \text{if } c < 1 \\ p_1, & \text{if } 1 \leq c < i \\ p_2, & \text{if } c \geq i \end{cases}$$

The number of contacts is cumulatively counted from the beginning of the simulation, rather than at the beginning of each time step. In other words, individuals only need to be exposed to $c \geq i$ different influential neighbors over

the course of the entire simulation instead of simultaneously at one time step to overcome i . This modeling choice was made because it may more realistically represent adoption behavior, such as that in ref. 21, especially when time of influence is short. Additionally, multiple exposures in one time step are still counted serially. If $i = 2$, and an individual is exposed to three influential neighbors for the first time in one time step, one neighbor “transmits” the likelihood of adoption with a rate of p_1 while the other two, and all subsequent exposures at later time steps with a rate of p_2 .

Seeding and Diffusion. At the start of the simulation, we seed the network with i individuals who have already adopted and are influential at the first time step. This is done to ensure that within the first time step, there is at least the possibility that a neighboring potential adopter will overcome the social reinforcement threshold and adopt with p_2 . For both clustered and random networks, one randomly chosen individual and a randomly selected set of $i - 1$ of its k neighbors are chosen as the original seeds. The seed individuals remain influential for T time steps. We consider the effects of alternative seeding strategies in [SI Appendix, section H](#).

Once an individual adopts a behavior, they remain influential for T time steps. After this, they are no longer influential, nor can they unadopt and then readopt. This is similar to being “recovered” or “removed” class in the SIR model (29). Individuals can also remain influential for the entire duration of the simulation after adopting (effectively having infinite T), similar to the SI model (29). Simulations run until a steady state is reached either when there are no more influential individuals, or no more susceptible individuals.

Simulation Trial Structure. For each parameter combination (p_1, p_2, k, T, i , and additional parameters covered in [SI Appendix](#)), the simulation is run on both the clustered and rewired regular random network 100 times. A different random network and starting seed set is used for each of the 100 trials. The clustered network remains unchanged from trial to trial because it is deterministically constructed and seeded.

The outcomes of interest include the reach of spread, measured as the proportion of individuals having adopted the behavior at the end of the simulation, and speed of spread, measured as the number of time steps required for 60 percent of the network to adopt a behavior (See [SI Appendix, section E](#) for a more detailed discussion). Both the amount of spread and speed outcomes are averaged over the 100 trials on each parameter combination.

Analytical Proof. We develop an analytical framework to determine the values of base rate p_1 and socially reinforced above threshold adoption rates p_2 for which full spread is likely to be reached as a function of network degree (k), social reinforcement threshold (i), and time of influence (T). To do so, we simplify the case to deriving the expected number of individuals, or “initial adopters” ($\langle I \rangle$), that will adopt from contact with the initial set of seed individuals in the first T time steps on either regular random or clustered ring lattice networks. Values of p_1 and p_2 where $\langle I \rangle$ equals or exceeds the number of initial seeds, which for our purposes is set equal to the social reinforcement threshold i , have a high chance of reaching full spread. Knowing exactly where $\langle I(p_1, p_2, k, i) = i \rangle$ creates a boundary between regions where full spread is either likely (when $\langle I \rangle \geq i$) or unlikely to be reached (when $\langle I \rangle < i$).

Since we are deriving expected values of initial adopters, there is still the possibility that full spread is not attained when $\langle I \rangle \geq i$ or when full spread is attained when $\langle I \rangle < i$. However, this probability is low considering that for most parameter combinations, the likelihood of a diffusion process reaching full spread is either really high or low ([SI Appendix, Fig. S6](#)). Qualitatively, this creates a visible boundary in the $p_1 \leq p_2$ space where full spread is highly unlikely to where full spread is highly likely. On random networks, $\langle I \rangle = i$ maps closely to the boundary between minimal and full spread observed in simulations. For clustered networks, $\langle I \rangle = i$ underestimates the boundary observed in simulations. Despite this imprecision, knowing where $\langle I \rangle = i$ in the $p_1 \leq p_2$ space allows us to accurately model how p_1, p_2, k, T , and i affect spread on random and clustered networks.

Finding $\langle I \rangle$ requires tracing the number of individuals who are exposed to at least one seed individual. Moreover, because adopting at either below or above threshold rates depends on the number of contacts to different seeds

a particular individual has, we must know the number of individuals having contact with exactly a seed individuals where at the least $a = 0$ and at most a is equal to the number of seeds i . We define j_a as the number of individuals having contact with exactly a seeds.

When the number of seeds an individual is exposed to is below the social reinforcement threshold ($a < i$), the independent probability of adopting at each contact is the base rate, p_1 . If seeds are only influential for one time step ($T = 1$), the cumulative probability of adoption with each successive contact where $a < i$ is $1 - (1 - p_1)^a$. If seeds are influential for multiple time steps ($T > 1$), this can be rewritten as $1 - (1 - p_1)^{Ta}$. Seeds may continuously influence their neighbors for each time step they are influential for, but only at the base rate p_1 .

When the number of seeds an individual is exposed to is equal to that of the social reinforcement threshold ($a = i$), when $T = 1$, the cumulative probability is $1 - (1 - p_1)^{i-1}(1 - p_2)$. Generalizing to cases where $T > 1$, this expression becomes $1 - (1 - p_1)^{i-1}(1 - p_2)^{T(i-1)}$. For the first $i - 1$ exposures to a seed, an individual adopts with rate p_1 . All subsequent exposures ($Ti - (i - 1)$) are adopted at the above threshold rate p_2 . Together, the cumulative probability of adoption, F_a can be expressed as

$$F_a = \begin{cases} 1 - (1 - p_1)^{Ta}, & \text{if } a < i \\ 1 - (1 - p_1)^{i-1}(1 - p_2)^{T(i-1)} & \text{if } a = i. \end{cases}$$

The sum of the product between the cumulative probability (F_a) and the number of individuals in contact with exactly a contacts (j_a) for each value of a yields the total expected number of initial adopters, $\langle I \rangle$.

$$\begin{aligned} \langle I \rangle &= \sum_{a=1}^i F_a j_a \\ &= \sum_{a=1}^{i-1} (1 - (1 - p_1)^{Ta}) j_a \\ &\quad + \sum_{a=i}^i (1 - (1 - p_1)^{i-1}(1 - p_2)^{T(i-1)}) j_a \\ &= \left[\sum_{a=1}^{i-1} (1 - (1 - p_1)^{Ta}) j_a \right] \\ &\quad + (1 - (1 - p_1)^{i-1}(1 - p_2)^{T(i-1)}) j_{a=i} \end{aligned}$$

Determining j_a , or the number of individuals in contact with exactly a seeds depends on the specific network structure.

Random networks. For analytical simplicity, we will assume random networks are seeded randomly rather than by choosing one individual and $i - 1$ neighbors as in the main results. Robustness checks show there is minimal difference in spread between these seeding strategies (SI Appendix, section H). We additionally assume that networks are sparse such that the degree is small relative to the number of individuals in the network, $k \ll n$. Under these assumptions, we show that as $n \rightarrow \infty$, $j_{a=1} \rightarrow ik$ and $j_{a>1} \rightarrow 0$. That is, there will be ik individuals that are connected to exactly one seed (k individuals for each seed) and no individuals connected to more than one seed.

There are two cases where there could be less than ik individuals that are connected to exactly one seed node. Either multiple seed nodes are adjacent to each other (case 1) or seed nodes share the same neighbor (case 2). We will show that the likelihood of either case occurring approaches zero as $n \rightarrow \infty$.

For case 1, as $n \rightarrow \infty$, each seed has k edges to possibly connect to one of the other $i - 1$ seeds. Since any given network is uniformly sampled from the space of all possible connected regular random networks, any individual has an equally likely chance to be connected to any other individual. This means any seed has k independent, equally likely chances of connecting to any of the other $n - 1$ individuals in the network. If there are $i - 1$ other seeds, there is a $(i - 1)/(n - 1)$ chance that one of the k edges of the target seed will connect to another seed. For large n and small numbers of seeds i , this can be modeled as a binomial draw where there are k chances for a seed node to connect to one of the

other $i - 1$ seeds with probability $(i - 1)/(n - 1)$, $\text{Bin}(k, \frac{i-1}{n-1})$. The expected number of neighbors of a seed that are themselves seeds is $k(i - 1)/(n - 1)$, which approaches 0 as $n \rightarrow \infty$.

For case 2, as $n \rightarrow \infty$, the probability that a nonseed neighbor of a seed is also connected to at least one other seed also approaches 0. Any neighbor of a seed has $k - 1$ edges that could connect to at least one of the $i - 1$ seeds out of $n - 2$ nodes (total number of nodes without the original seed and seed neighbor). By the same logic as case 1, this can be modeled as a binomial draw where a node has $k - 1$ chances of selecting one of the other $i - 1$ seeds with probability $(i - 1)/(n - 2)$, $\text{Bin}(k - 1, \frac{i-1}{n-2})$. The expected number of seeds this neighbor of the original seed is connected to is $(k - 1)(i - 1)/(n - 2)$, which also tends toward 0 as $n \rightarrow \infty$.

In sparse networks, any seed is unlikely to be neighbors with a seed or to share neighbors with another seed. This means each of the i seeds are connected to k unique neighbors, such that there are ik total individuals connected to exactly one seed and no individuals connected to more than one seed. The expected number of initial adopters will be that where $j_{a=1} = ik$ and $j_{a>1} = 0$. Since $i > 1$, $j_{a=i} = 0$.

$$\begin{aligned} \langle I_R \rangle &= \sum_{a=1}^i F_a j_a \\ &= \left[\sum_{a=1}^{i-1} (1 - (1 - p_1)^{Ta}) j_a \right] \\ &\quad + (1 - (1 - p_1)^{i-1}(1 - p_2)^{T(i-1)}) j_{a=i} \\ &= (1 - (1 - p_1)^{T(1)}) (j_{a=1} + j_{a=2} \dots j_{a=i-1}) \\ &\quad + (1 - (1 - p_1)^{a-i}(1 - p_2)^{T(i-1)}) j_{a=i} \\ &= (1 - (1 - p_1)^{T(1)}) (ik + 0 \dots 0) \\ &\quad + (1 - (1 - p_1)^{a-i}(1 - p_2)^{T(i-1)}) (0) \\ &= (1 - (1 - p_1)^T) ik \end{aligned}$$

To continue the diffusion process, in expectation there must be one initial adopter for each seed. This logic is similar to the epidemic threshold (29). Within the first T time steps the seeds are influential for, the i seeds must transmit the behavior to at least i other individuals to continue the diffusion process, so

$$\begin{aligned} \langle I_R \rangle &= (1 - (1 - p_1)^T) ik \geq i \\ 1 - (1 - p_1)^T &\geq \frac{i}{ik} \\ 1 - (1 - p_1)^T &\geq \frac{1}{k} \end{aligned}$$

Beyond the initial seed nodes, an influential individual must influence at least one neighbor out of their $k - 1$ neighbors who have not yet adopted (assuming one neighbor has already adopted in order to pass on the behavior to the currently influential individual). This means $1 - (1 - p_1)^T \geq 1/(k - 1)$, from which we calculate p_1^* , or the value that the base rate p_1 must be equal or exceed in order for diffusion to be sustained on random networks.

$$\begin{aligned} p_1 &\geq 1 - \left(1 - \frac{1}{(k - 1)} \right)^{1/T} \\ p_1^* &= 1 - \left(1 - \frac{1}{(k - 1)} \right)^{1/T} \end{aligned}$$

The equation for p_1^* shows that spread on random networks only depends on p_1 , k , and T .

Clustered networks. In clustered networks, as a simplifying assumption, we pick i adjacent individuals as the initial seeds instead of one randomly selected individual and $i - 1$ randomly selected neighbors. We observe minimal

differences between these two seeding strategies (*SI Appendix, Fig. S8*). Specific to selecting adjacent seeds,

$$j_a = \begin{cases} 2, & \text{if } a < i \\ 2(\frac{k}{2} - i + 1), & \text{if } a = i \end{cases}$$

By the same logic as the random network, the overall number of initial adopters can be expressed as

$$\langle I_L \rangle = 2 \sum_{a=1}^{i-1} (1 - (1 - p_1)^{T a}) + (1 - (1 - p_1)^{i-1} (1 - p_2)^{T i + 1}) (k - 2i + 2)$$

Setting $\langle I_L \rangle = i$ and solving for p_2 ,

$$p_2 = \frac{2 \sum_{a=1}^{i-1} (1 - (1 - p_1)^{T a}) - 3i + k + 2}{(k - 2i + 2)(1 - p_1)^{i-1}}$$

While the $\langle I_L \rangle = i$ boundary accurately models the regions of the $p_1 \leq p_2$ space for which the average number of initial adopters equals or exceeds the

number of seeds ($\langle I_L \rangle \geq i$), it underestimates the values of p_1 and p_2 for which full spread is likely reached on clustered networks (*SI Appendix, Fig. S1*). Despite this discrepancy, the effect of k , i , and T on the $\langle I_L \rangle \geq i$ boundary, mirrors the effect of these parameters on the simulated results. That is, as k and T increase, diffusion on clustered networks becomes more likely at increasingly lower values of p_1 and p_2 . As i increases, diffusion becomes less likely as contact to a greater number of influential neighbors are required for an individual to adopt with the higher p_2 rate.

Data, Materials, and Software Availability. Code data have been deposited in OSF (https://osf.io/8jfcy/?view_only=dc9a6a759be941caa618621e2974a83b) (24).

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